

Math 465: Introduction to Combinatorics

Yelena Mandelshtam

Homework #2 due tonight by 11:59PM.

Quiz #2 will be on Thursday in class.

Course policy update: “I reserve the right to ask any student to meet briefly to explain or present a solution they submitted on a homework assignment or quiz. Such a meeting may be used to verify the student’s understanding of the submitted work and, when appropriate, its authorship. You should be prepared to explain the reasoning behind any work you submit.”

These slides will be posted on Canvas.

Next topic: applications of the Multiplication Principle for generating functions to **refined enumeration of permutations**.

Definition

We denote by S_n the set of permutations of $\{1, \dots, n\}$. For example,

$$S_3 = \{123, 132, 213, 231, 312, 321\}.$$

A weight function on permutations is called a **permutation statistic**.

Inversions

Definition

We denote by $\text{inv}(w)$ the number of **inversions** in a permutation w .

Example

$$w = 35142$$

$$3\,5\,1\,4\,2 \quad 3\,5\,1\,4\,2 \quad 3\,5\,1\,4\,2 \quad 3\,5\,1\,4\,2 \quad 3\,5\,1\,4\,2 \quad 3\,5\,1\,4\,2$$

$$\text{inv}(w) = 6$$

Example: $n = 3$

w	123	132	213	231	312	321
$\text{inv}(w)$	0	1	1	2	2	3

The corresponding generating function is

$$\sum_{w \in S_3} x^{\text{inv}(w)} = 1 + 2x + 2x^2 + x^3 = (1+x)(1+x+x^2).$$

Counting permutations by the number of inversions

Theorem

The generating function for counting permutations in S_n with respect to the number of inversions is given by

$$\begin{aligned}\sum_{w \in S_n} x^{\text{inv}(w)} &= 1 \cdot (1 + x) \cdot (1 + x + x^2) \cdots (1 + \cdots + x^{n-1}) \\ &= \prod_{k=1}^n \frac{1 - x^k}{1 - x}.\end{aligned}$$

The proof of this theorem will rely on the notion of the [code](#) of a permutation.

Code of a permutation

The **code** of a permutation $w \in S_n$ is the sequence $(c_1, \dots, c_n)$ where $c_k = \#\{\text{entries to the right of } k \text{ which are } < k\}$.

In other words: place 1, 2, 3, ..., one by one, and record the number of inversions each new entry creates:

w	$\text{inv}(w)$	$\text{code}(w)$
123	0	(0, 0, 0)
132	1	(0, 0, 1)
213	1	(0, 1, 0)
231	2	(0, 1, 1)
312	2	(0, 0, 2)
321	3	(0, 1, 2)

Observations

- $\text{inv}(w) = c_1 + \dots + c_n$;
- each c_k takes values $0, 1, \dots, k-1$.

Counting permutations by the number of inversions

The bijection

$$\begin{aligned} S_n &\rightarrow \{0\} \times \{0, 1\} \times \{0, 1, 2\} \times \cdots \times \{0, 1, \dots, n-1\} \\ w &\mapsto (c_1, c_2, c_3, \dots) \end{aligned}$$

satisfies the additivity condition if we assume that

- the weight of w is $\text{inv}(w)$,
- the weight of each entry c_i is c_i .

Applying the Multiplication Principle for generating functions, we get:

Theorem

The generating function for counting permutations in S_n with respect to the number of inversions is given by

$$\sum_{w \in S_n} x^{\text{inv}(w)} = 1 \cdot (1 + x)(1 + x + x^2) \cdots (1 + \cdots + x^{n-1}).$$

Cycle notation

$$w = 5276134$$



$$w = (15)(2)(3746)$$

Example: $n = 3$

w	cycle notation
123	$(1)(2)(3)$
132	$(1)(23)$
213	$(12)(3)$
231	(123)
312	(132)
321	$(13)(2)$

Counting cycles

Definition

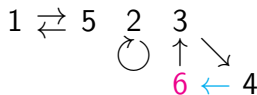
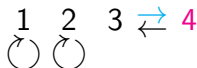
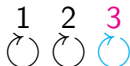
Let $c(w)$ denote the number of cycles in a permutation w .

Example

w	cycle notation	$c(w)$
123	(1)(2)(3)	3
132	(1)(23)	2
213	(12)(3)	2
231	(123)	1
312	(132)	1
321	(13)(2)	2

$$\sum_{w \in S_3} x^{c(w)} = x^3 + 3x^2 + 2x = x(x+1)(x+2)$$

Building a permutation by inserting entries into cycles



$$\text{encoding} = (b_1, \dots, b_7) = (0, 0, 0, 3, 1, 4, 3)$$

The k th entry of the **encoding** can take values $0, 1, 2, \dots, k-1$.

Encoding of a permutation vs. the number of cycles

w	cycle notation	$c(w)$	encoding $(b_1, \dots, b_n)$
123	(1)(2)(3)	3	(0, 0, 0)
132	(1)(23)	2	(0, 0, 2)
213	(12)(3)	2	(0, 1, 0)
231	(123)	1	(0, 1, 2)
312	(132)	1	(0, 1, 1)
321	(13)(2)	2	(0, 0, 1)

$$\begin{aligned}c(w) &= \text{number of 0's in the encoding of } w \\ &= \beta(b_1) + \dots + \beta(b_n)\end{aligned}$$

$$\beta(b) \stackrel{\text{def}}{=} \begin{cases} 1 & \text{if } b = 0; \\ 0 & \text{otherwise.} \end{cases}$$

Counting permutations by the number of cycles

Theorem

The generating function for counting permutations in S_n with respect to the number of cycles is given by

$$\sum_{w \in S_n} x^{c(w)} = x(x+1)(x+2) \cdots (x+n-1).$$

Proof

Each permutation $w \in S_n$ is encoded by a sequence $(b_1, \dots, b_n)$. The additivity condition is satisfied:

$$c(w) = \beta(b_1) + \cdots + \beta(b_n).$$

The generating function for each b_k is $x + k - 1$ [why?].

By the Multiplication Principle for generating functions, we obtain

$$\sum_{w \in S_n} x^{c(w)} = \prod_k (x + k - 1).$$

Signless Stirling numbers of the first kind

Recall that $c(w)$ denotes the number of cycles in a permutation w .

Definition

The numbers

$$c(n, k) = \#\{w \in S_n \mid c(w) = k\} \quad (1 \leq k \leq n)$$

are called the signless Stirling numbers of the first kind.

There is no explicit closed formula for $c(n, k)$.

Corollary

$$\sum_{k=1}^n c(n, k)x^k = \sum_{w \in S_n} x^{c(w)} = x(x+1)(x+2) \cdots (x+n-1)$$

Recurrence for the signless Stirling numbers of the 1st kind

Proposition

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$$

Proof

The formula

$$\sum_{k=1}^n c(n, k)x^k = x(x+1)(x+2)\cdots(x+n-1)$$

implies the identity

$$\sum_{k=1}^n c(n, k)x^k = (x+n-1) \sum_{j=1}^{n-1} c(n-1, j)x^j.$$

Now compare the coefficients of x^k on both sides.

Exercise

Give a combinatorial proof of this proposition.

Triangle of the signless Stirling numbers of the first kind

The recurrence

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$$

can be used to recursively compute the Stirling numbers $c(n, k)$.

Triangle of the numbers $c(n, k)$ for $1 \leq k \leq n \leq 7$

							1						
						1		1					
				2			3		1				
		6			11		6			1			
	24		50			35		10			1		
120		274		225		85		15				1	
720		1764		1624		735		175		21			1
...	...	...	...	...	...	...	...	...	...	...	...	...	...

The numbering of rows/columns starts with $n = 1$ (resp., $k = 1$).

Signed Stirling numbers of the first kind

Definition

The numbers

$$s(n, k) = (-1)^{n-k} c(n, k)$$

are called the **signed Stirling numbers of the first kind**.

				1						
				-1		1				
		2			-3		1			
		-6		11		-6		1		
	24		-50		35		-10		1	
-120		274		-225		85		-15		1
...	...	...	...	...	...	...	...	...	...	...

Signed Stirling numbers of the first kind $s(n, k)$ for $n \leq 6$